



In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CE G613
Course Title : **ADVANCED CONCRETE STRUCTURES**
Instructor-in-Charge : G MUTHUKUMAR

1. Course Description: The course deals with various aspects of concrete structures with emphasis on design of various elements and exposure to different code provisions.

Materials; high strength concrete, flexure analysis and design; shear and diagonal tension; bond and anchorage; serviceability; torsion; columns; joints; indeterminate beams and frames; yield line analysis; strip method for slabs; composite construction; footing and foundations; concrete building system; concrete tall buildings, detailing in concrete structures

2. Scope and Objective of the Course: The concrete structures are quite common in India for both low-rise and high-rise buildings due to the strong performance in the last many decades even during strong wind & earthquake ground motion. The concrete structures are used in most of the infrastructure applications such as construction of buildings, bridges, roads, airport structures, stadiums, water tanks, chimneys, storage bins, nuclear reactors etc., The codal provisions are continuously changing with changing times due to technological advancements and our improved understanding of the behavioral characteristics of concrete. Though concrete structures are design are not anything new, but certain aspects of design and construction still needs further research and understanding. The beam-column joint is hardly 50 years old and was totally unexplored before 1970s. Today, it was observed that beam-column failure is one of the most important reasons for the failure of RC framed structures. On the other hand, RC structures are only framed form, but different structural configurations evolve over the years to accommodate the tall buildings with safety and economy. The present course aims to highlight the design and construction aspects of wide range of structures, ranging from traditional RC framed structures to tall buildings to certain specialized structures such as chimneys and water tanks. The emphasis will be more on the performance of structure under earthquake loading conditions. The codal provisions will be discussed for the better understanding.

- To understand the concrete structures and its applications [**Module - I**]
- To study different structural configurations for the concrete structures [**Module-II**]
- To understand and design few important components of RC framed structures including beam-column joints and other vital structural elements [**Module-III**]
- To understand the concept of shear wall frame structures [**Module IV**]
- To understand the behavioral and design aspects of shell structures and folded plates [**Module V**]
- To understand the ductile behavioral aspects of structures under earthquake [**Module VI**]
- To understand the design and detailing aspects of water tanks and chimneys [**Module VII**]
- To understand the design aspects of RCC poles and pipes [**Module VIII**]



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- To discuss the case studies of different RC structures and its failures [**Module IX**]
- To discuss the various advancements in concrete structures and the research scope [**Module X**]

3. Course learning outcomes:

- Comprehensive understanding of design aspects of concrete structures
- Structural elements and code provisions
- RC detailing of critical structural elements
- Tall buildings behaviour and structural systems

4. Text Books & Reference Books:

T1. P.C. Varghese, "Advanced Reinforced Concrete Design", second edition, PHI Learning Pvt. Ltd., New Delhi.

T2. S. U. Pillai and D. Menon, "Reinforced Concrete Design", Third Edition, Tata McGraw Hill.

- R1: H.J. Shah, "Reinforced concrete Vol. II (Advanced Reinforced Concrete)", Seventh edition, Charotar Publishing House Pvt. Ltd., Anand.
- R2: B.S. Taranath, "Reinforced Concrete Design of Tall Buildings", Special Indian Edition, CRC Press, New York.
- R3: IS 456, IS 875, IS 1893, IS 3370, IS 6533, IS 13920.
- R4: S. N. Manohar, "Tall chimneys: Design and construction", Tata McGraw Hill.
- R5: Raju, N.K., "Advanced RCC Design", CBS Publishers and distributors, New Delhi.
- R6: Gambhir, M.L., "Design of Reinforced Concrete Structures", PHI Learning Pvt. Ltd., New Delhi.
- R7: Smith and Coull, "Tall Buildings Structures: Analysis and design", Wiley Publications.

5. Method of conducting the course:

- Both blackboard and PPT will be used during the lecture session
- Lecture notes will be uploaded on Nalanda for reference

6. Course Plan:

Mod. No.	Lecture No.	Module	Lecture session	Reference	Learning outcomes
1	1-2	Concrete structures and its application	Advantages of concrete structures over steel structures; History; Characteristics; Present state-of-the art	T2, Class notes	Key characteristics of concrete and steel structures; Past records
2	3-6	Structural configurations	RC framed structures; Shear wall-Frame structures, Tubular structures (Framed tube, Braced frame tube, tube-in-tube structures)	T1,T2, Class notes	Different structural configurations and their advantages, failure modes. Tall building scenario.
3	7-11	RC framed structures	Design aspects of framed structures (including slab, flexural, shear	T1, T2, Class notes	Ductile frames structures; capacity



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			behavior of beams); Design of beam-column joints; Design of deep beams; Yield line analysis of slab, flat slab, composite construction		of the joint; flexure and shear design; slender and deep beams; Side face reinforcements;
4	12-14	Concept of ductility & dynamic ground motion	Earthquake Resistant Design; Earthquake ground motion; Concept of storey drift; Accidental eccentricity; Ductile design considerations; Strength hierarchy; Strong column weak beam philosophy; Special confining reinforcement; Footing and Foundations	R3	Design aspects of important components of building. Rationale behind the moment capacity code specifications, footing and foundation system
4	15-19	Shear wall-frame structures	Behavior of shear wall frame interaction; Analysis of shear-wall frame structures; Design aspects of shear wall with and without boundary elements	R2,R7	Design aspects and detailing practices of shear wall, Spacing, geometry and anchorage etc. Frame shear wall interaction advantage; approximate analysis; special confinement at boundary element columns;
5	20-24	Shell structures and folded plates	General classification of shells; Design aspects of Shell roofs; Hyperbolic cooling towers; Spherical domes; folded plates;	R1,R6, JOURNAL ARTICLES	Advantages of shells and folded plates; thickness requirements; load considerations; spherical domes; roof slabs;
7	25-28	Water tanks and chimneys	Seismic analysis and design aspects of overhead water tank (Rectangular and Intze type) (IS: 3370-2009); Design aspects of chimney subjected to lateral loads; [R2 and R5]	R1,R4	Water tank methods of design; code provisions and key aspects; Reinforcement detailing, spacing, meridional and hoop stresses; capacity determination; ring beam, raft foundations and components; dowel bar, anchorage;
8	29-31	RCC Poles and Pipes	Specifications for overhead power and telecommunication lines; RCC Pipes; Concrete pipe design; Special considerations; Buoyancy forces; Pipe-soil structural interaction;	[Journal articles and class notes]	Design aspects of self supporting and guyed towers; Use of nonlinear analysis; seismic consideration; Height, mass stiffness considerations;



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					recent research; Structural Standard for Antenna Supporting Structures and Antennas;
9	32-37	RC structures and its failures	Failure modes of RC structures; Poor construction practices; Stability of RC structures; Common design errors; Unavoidable complexities in construction	[Class notes]	Reasons behind failures during construction; Importance of design of temporary structures; loading during construction;
10	38-40	Advancements in concrete structures;	Introduction to Performance Based design; New structural configurations; Structural design software; [Class notes]	Class notes	Performance based designs using commercial softwares; recent innovations in configurations and stringent requirements; International best practices;
			Total	40	

7. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Comments
Announced Class Quiz 1	5%	50 minutes (each)	August 24 th , 2026 (Class hour)	Open book
Announced Class Quiz 2	5%	50 minutes (each)	September 23 rd , 2026 (Class hour)	Open book
Announced Class Quiz 3	5%	50 minutes (each)	November 16 th , 2026 (Class hour)	Open book
Course Project	25%	1 month	November 15 th , 2026	Open book
Mid-Semester Examination	25%	1.5 hours	<TEST_1>	Open Book
Comprehensive Examination	35%	3 hours	<TEST_C>	Closed book/Open Book



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Criterion for NC: If the candidate misses one particular component completely, NC will be awarded. Also, if the candidate is appearing for the sake of appearing, NC will be awarded.

8. Chamber Consultation Hour: Thursday (6 PM – 7 PM).

9. Make-up Policy: Make-up will be granted only to genuine cases. For cases related to illness, proper documentary evidence is essential.

10. Note (if any): Students are expected to attend the classes sincerely. Notice, if any, concerning the course will be displayed on the Nalanda LMS.

Instructor-in-charge
CE G613